

On the Feasibility of a Censorship Resistant Decentralized Name System

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Where We Are



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Adversary Model

- ▶ Any role
- ▶ Multiple Identities
- ▶ Computational Power
- ▶ Legal Power

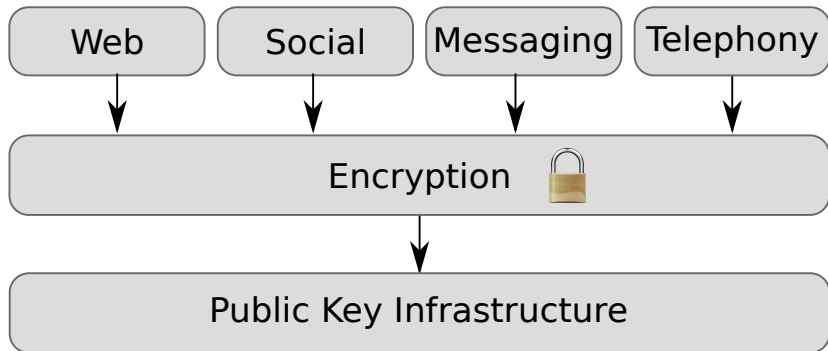


But cannot:

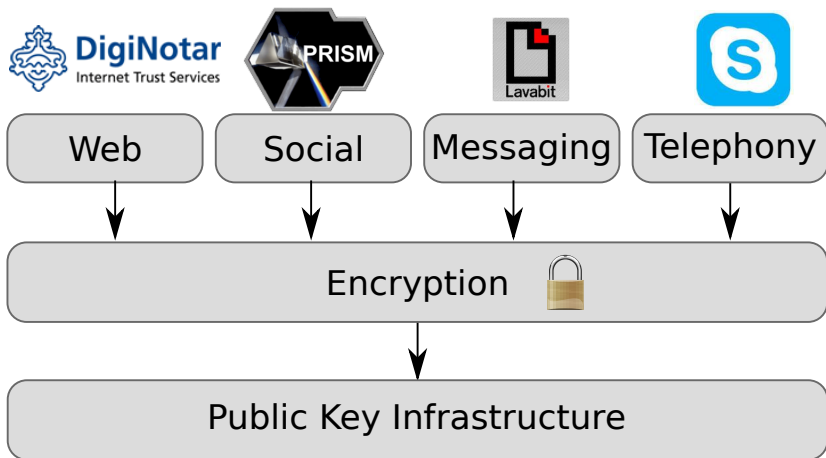
- ▶ Break or prevent crypto
- ▶ Attack user systems
- ▶ Prevent network communication



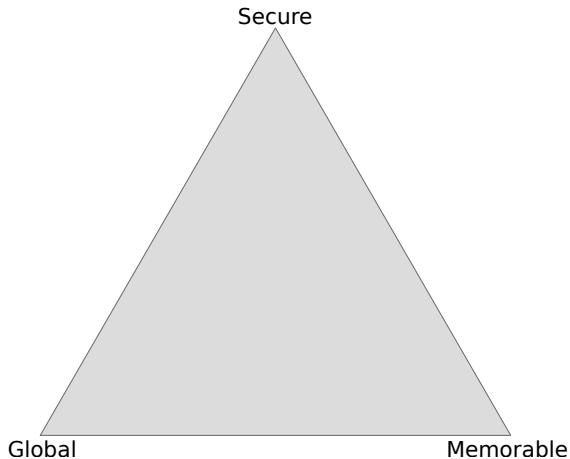
Broken Pillars



Broken Pillars

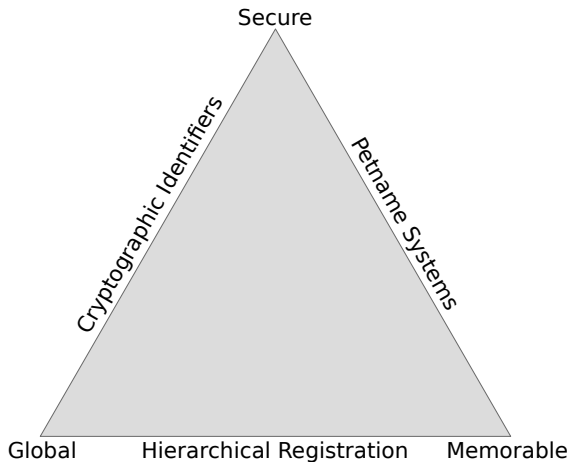


Zooko's Triangle

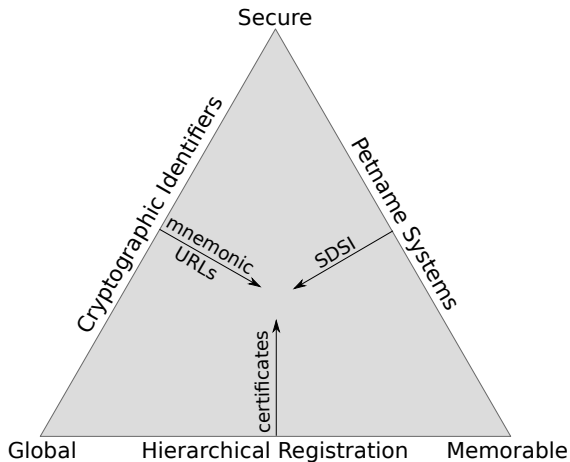


A name system can only fulfill **two**!

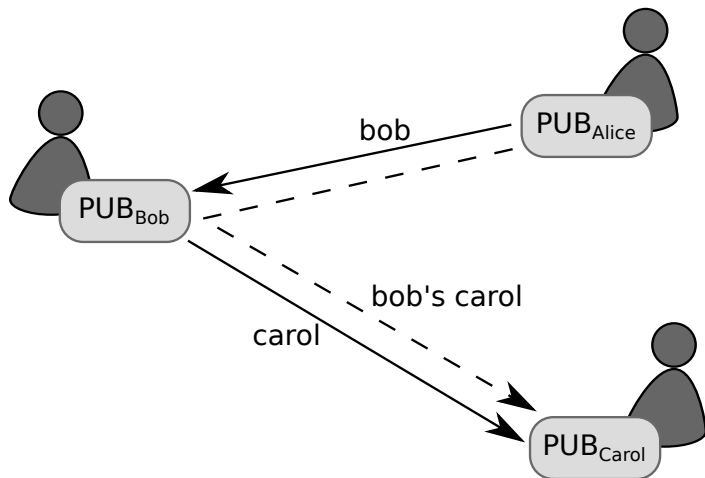
Zooko's Triangle



Zooko's Triangle



Petnames and SDSI



Requirements

- ▶ Censorship Resistance by Decentralization
⇒ Peer-to-Peer, DHT
- ▶ Verification
⇒ Signatures
- ▶ Query Privacy
⇒ Encryption
- ▶ Integration with DNS
⇒ Coexisting Namespaces

Challenges

- ▶ Legacy Applications
⇒ Proxies
- ▶ Censorship in other Layers
⇒ Tor
- ▶ Usability
⇒ DNS look-a-like, QR Code



End

Thank you!

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Get the code:

<https://gnunet.org/gns>